

package- collection of module

- module is collection of function
- function - is collection of statements
- in build functions ---print(),max(),min(),len(),id()
- user defined function always assigned with def keyword

NUMPY

```
In [1]: import numpy as np
```

```
In [5]: np.__version__
```

```
Out[5]: '2.3.5'
```

```
In [6]: import sys  
sys.version
```

```
Out[6]: '3.13.9 | packaged by Anaconda, Inc. | (main, Oct 21 2025, 19:09:58) [MSC v.1929 6  
4 bit (AMD64)]'
```

creating arrays

```
In [7]: my_list = [0,1,2,3,4,5]  
my_list
```

```
Out[7]: [0, 1, 2, 3, 4, 5]
```

```
In [8]: type(my_list)
```

```
Out[8]: list
```

```
In [9]: # List converting to array
```

```
In [11]: arr = np.array(my_list)
```

```
In [12]: arr
```

```
Out[12]: array([0, 1, 2, 3, 4, 5])
```

```
In [13]: type(arr)
```

```
Out[13]: numpy.ndarray
```

```
In [14]: type(my_list)
```

Out[14]: list

In [17]: `np.arange(10)# print`

Out[17]: array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])

In [18]: `np.arange(3.0)# float output`

Out[18]: array([0., 1., 2.])

In [19]: `np.arange(0,5)`

Out[19]: array([0, 1, 2, 3, 4])

In [20]: `np.arange(10,20)`

Out[20]: array([10, 11, 12, 13, 14, 15, 16, 17, 18, 19])

In [21]: `np.arange(20,10) # first arg > second arg`

Out[21]: array([], dtype=int64)

In [22]: `np.arange(-20,-10)`

Out[22]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11])

In [23]: `np.arange()`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[23], line 1  
----> 1 np.arange()  
  
TypeError: arange() requires stop to be specified.
```

In [24]: `np.arange(10,30,5)`

Out[24]: array([10, 15, 20, 25])

In [25]: `np.arange(0,10,2)`

Out[25]: array([0, 2, 4, 6, 8])

In [26]: `np.arange(10,20,30,40) # only three arguments`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[26], line 1  
----> 1 np.arange(10,20,30,40)  
  
TypeError: Cannot interpret '40' as a data type
```

In [28]: `np.zeros(3)# paramater tunning`

```
Out[28]: array([0., 0., 0.])
```

```
In [29]: np.zeros(10)# output- float to convert to int hyperparameter tuning
```

```
Out[29]: array([0., 0., 0., 0., 0., 0., 0., 0., 0., 0.])
```

```
In [31]: np.zeros(10,dtype=int) # hyper parameter tuning
```

```
Out[31]: array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0])
```

```
In [32]: # parameter tuning- system by default  
#hyper parameter tuning - user
```

```
Cell In[32], line 2  
    hyper parameter tuning - user  
      ^  
SyntaxError: invalid syntax
```

```
In [35]: np.zeros((2,2),dtype=int) # 2d dimension array
```

```
Out[35]: array([[0, 0],  
               [0, 0]])
```

```
In [37]: np.zeros((2,2))
```

```
Out[37]: array([[0., 0.],  
               [0., 0.]])
```

```
In [38]: np.zeros((2,10))
```

```
Out[38]: array([[0., 0., 0., 0., 0., 0., 0., 0., 0., 0.],  
               [0., 0., 0., 0., 0., 0., 0., 0., 0., 0.]])
```

```
In [41]: np.zeros((5,10),dtype=int)
```

```
Out[41]: array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],  
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

```
In [42]: np.ones(3)
```

```
Out[42]: array([1., 1., 1.])
```

```
In [43]: np.ones(3, dtype=int)
```

```
Out[43]: array([1, 1, 1])
```

```
In [50]: np.ones((5,4))
```

```
Out[50]: array([[1., 1., 1., 1.],
               [1., 1., 1., 1.],
               [1., 1., 1., 1.],
               [1., 1., 1., 1.],
               [1., 1., 1., 1.]])
```

```
In [52]: random.rand(3,2)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[52], line 1
----> 1 random.rand(3,2)

NameError: name 'random' is not defined
```

```
In [53]: rand(3,5)
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[53], line 1
----> 1 rand(3,5)

NameError: name 'rand' is not defined
```

```
In [54]: np.random.rand(3,6)
```

```
Out[54]: array([[0.69933434, 0.07926309, 0.70919369, 0.14514619, 0.50250145,
                 0.79372183],
               [0.31287746, 0.17192807, 0.36697403, 0.24590589, 0.12214231,
                 0.99999112],
               [0.39103523, 0.96490641, 0.95724993, 0.4162502 , 0.65524449,
                 0.60810831]])
```

```
In [55]: np.rand(4) # module missing
```

```
-----
AttributeError                            Traceback (most recent call last)
Cell In[55], line 1
----> 1 np.rand(4)

File ~\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'rand'
```

```
In [56]: np.random.rand(3,5)
```

```
Out[56]: array([[0.99120201, 0.72913606, 0.17534308, 0.09413439, 0.44342373],
               [0.58166328, 0.06913389, 0.68816738, 0.73736056, 0.19463787],
               [0.43382429, 0.39264379, 0.87626521, 0.44945363, 0.90925631]])
```

```
In [59]: np.random.randint(2,20) # 2 inclusive 20 exclusive
```

```
Out[59]: 13
```

```
In [60]: np.random.randint(10,20,3)
```

```
Out[60]: array([15, 16, 16], dtype=int32)
```

```
In [74]: np.arange(1,10).reshape(12,1)
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[74], line 1  
----> 1 np.arange(1,10).reshape(12,1)  
  
ValueError: cannot reshape array of size 9 into shape (12,1)
```

```
In [ ]:
```